

7.5 $l_0 = 0.5 \text{ m}$, $m = 0.1 \text{ kg}$, $l = 0.6 \text{ m}$, $g = 9.8 \text{ m/s}^2$, $\Delta l = l - l_0 = 0.1 \text{ m}$

(1) 平衡位置, $mg = k\Delta l$, 得 $k = \frac{mg}{\Delta l} = 9.8 \text{ N/m}$.

↓
以 为原点, 向下为正方向, 位移为 x 时, 所受合力

$$F = mg - k(l+x) = mg - k\Delta l - kx$$

$$\text{由 } mg = k\Delta l \Rightarrow F = -kx \text{ 为简谐振动}$$

(2) $A = \Delta l = 0.1 \text{ m}$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{98} \approx 9.9 \text{ rad/s}, f = \frac{\omega}{2\pi} = 1.58 \text{ Hz}.$$

(3) 设 $x = A \cos(\omega t + \varphi)$. $t = 0$ 时, $x_0 = -0.1 \text{ m}$, 又 $A = 0.1 \text{ m}$

$$\Rightarrow \cos \varphi = -1 \Rightarrow \varphi = \pi$$

$$\Rightarrow x = 0.1 \cos(9.9t + \pi)$$

2.7

$$A = 0.1 \text{ m}, \lambda_1 = 0.06 \text{ m}, v_1 = 0.24 \text{ m/s}$$

$$(1) \text{ 由 } v_1^2 = \omega^2 (A^2 - x_1^2)$$

$$\text{得 } \omega = 3 \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3} \text{ s} \approx 2.09 \text{ s}$$

$$(2) v_2^2 = \omega^2 (A^2 - x_2^2)$$

$$\Rightarrow x_2 = \pm \sqrt{0.0084} \text{ m} \approx \pm 9.17 \text{ cm}$$

2.15 (1) 设底面积 S , 水密度 ρ , 平衡时浸入高度为 a .

阿基米德原理: $G = F_{\text{浮}} = \rho g S a$ (平衡)

以平衡位置为原点, 竖直向下为正方向:

$$F_{\text{合}} = G - F_{\text{浮}} = G - \rho g S (a + x) = -\rho g S x$$

$$\text{符合 } F_{\text{合}} = -kx, k = \rho g S$$

故为简谐运动

$$(2) m = \frac{G}{g} = \rho S a$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{\rho S a}{\rho g S}} = 2\pi \sqrt{\frac{a}{g}}$$

$$(3) A = |b - a| \text{ (相对平衡位置的最大位移即振幅)}$$

7.17 圆盘绕悬挂轴

的转动惯量: $I = I_c + Md^2 = \frac{1}{2}MR^2 + Md^2$

圆盘偏离 θ 时, 重力矩 $M_z = -Mgd\sin\theta$,

又 $\sin\theta \approx \theta$ (小角度),

$$\textcircled{*} \quad M_z = I\alpha = I \frac{d^2\theta}{dt^2}$$

$$\Rightarrow \frac{d^2\theta}{dt^2} + \frac{Mgd}{I} \theta = 0$$

~~$$\frac{1}{2} \omega^2 = \frac{Mgd}{I} \theta(t) = \omega^2 t$$~~

~~$$\Rightarrow \frac{1}{2} \omega^2 + \omega^2 = 0$$~~

~~$$\Rightarrow r = \pm t\omega$$~~

~~$$\theta(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t)$$~~

$$\Rightarrow \omega = \sqrt{\frac{Mgd}{I}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{Mgd}}$$

7.25 $I = \frac{1}{3}ml^2$ (细杆对端点).

偏离一个小角度 θ 后,

$$M_g = -mg \cdot \frac{l}{2} \sin\theta \approx -\frac{1}{2}mg l \theta.$$

弹簧弹力矩 M_k :

$$\Delta x = l\theta, F = kl\theta,$$

$$M_k = -Fl \cos\theta \approx -kl^2\theta \quad (\cos\theta \approx 1)$$

$$\text{故由 } M = I\alpha = I \frac{d^2\theta}{dt^2}$$

$$\Rightarrow \frac{1}{3}ml^2 \frac{d^2\theta}{dt^2} = M_g + M_k$$

$$\frac{1}{3}ml^2 \frac{d^2\theta}{dt^2} + \frac{1}{2}mg l \theta + kl^2\theta = 0$$

$$\Rightarrow \frac{d^2\theta}{dt^2} + \frac{3g + \frac{6kl}{m}}{2l} \theta = 0$$

$$\Rightarrow \omega = \sqrt{\frac{3g + \frac{6kl}{m}}{2l}} = \sqrt{\frac{3g}{2l} + \frac{3k}{m}}$$

$$\Rightarrow f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{3g}{2l} + \frac{3k}{m}}$$

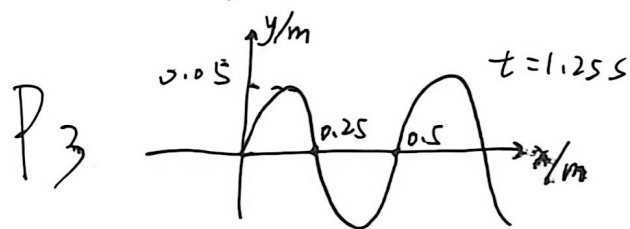
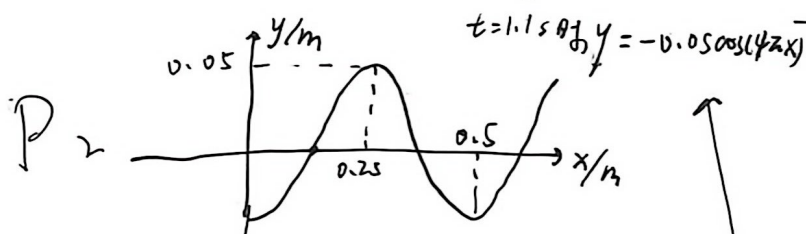
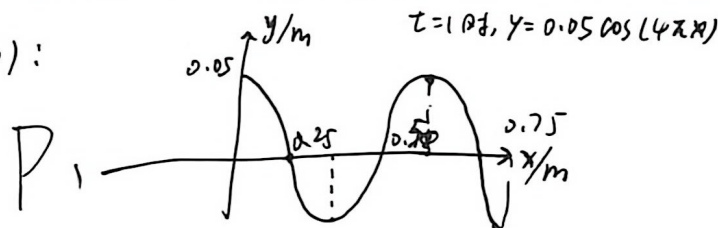
13.12 $y = 6 \times 10^{-2} \cos \frac{\pi t}{5} \text{ (m)}, u = 20 \text{ m/s},$

(1) $t = \frac{x}{u} = 3 \text{ s},$

波源 3 s $\Rightarrow y = 6 \times 10^{-2} \cos \frac{\pi}{5} (t-3)$
 $= 6 \times 10^{-2} \cos (\frac{2\pi t}{5} - \frac{3\pi}{5}) \text{ cm}.$

(2) $\Delta \phi = \omega \frac{x}{u} = \frac{3\pi}{5} = 0.6\pi \text{ rad}.$

13.14 (3):



$t = 1.5 \text{ s}$
 $y = -0.05 \cos(4\pi x)$

与 P_2 重合

13.14. $y = 0.05 \cos(10\pi t - 4\pi x).$

(1) 由 $y = A \cos(2\pi f t - \frac{2\pi}{\lambda} x)$ 对比

$\Rightarrow \lambda = 0.5 \text{ m}, f = 5 \text{ Hz}, u = \lambda f = 2.5 \text{ m/s}$

(2) $x = 0.2 \text{ m}, t = 1 \text{ s};$

$\phi = 10\pi \cdot 1 - 4\pi \cdot 0.2 = 9.2\pi \text{ rad}$

等效于 $\phi' = \phi - 4 \times 2\pi = 1.2\pi \text{ rad}.$

或 $\phi - 5 \times 2\pi = -0.8\pi \text{ rad}.$

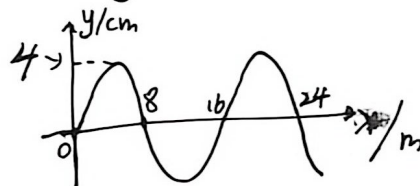
对应 $t' = t - \frac{x}{u} = 0.92 \text{ s}$ 处相位.

13.16 $u = 4 \text{ m/s}, A = 4 \text{ cm}, T = 4 \text{ s}.$

(1) $\omega = \frac{2\pi}{T} = \frac{\pi}{2} \text{ rad/s}, t = 0 \text{ s}, y = 4 \text{ cm}, \phi_0 = 0$

$y = A \cos[\omega(t - \frac{x}{u})] = 4 \times 10^{-2} \cos \frac{\pi}{2} (t - \frac{x}{4}) \text{ m}.$

(2) $t = 3 \text{ s}, y = \cos(\frac{3\pi}{2} - \frac{\pi}{8} x) = \sin(\frac{\pi x}{8}) \text{ cm}$



$T = \frac{2\pi}{\frac{\pi}{8}} = 16 \text{ s}.$

13.33.

$$f_s = 1 \text{ kHz} = 1000 \text{ Hz}, v_s = 10 \text{ m/s}, u = 330 \text{ m/s},$$

$$(1) f_1 = f_s \frac{u}{u + v_s} = 1000 \times \frac{330}{340} \approx 970.6 \text{ Hz}$$

$$(2) f_2 = f_s \frac{u}{u - v_s} \approx 1031.3 \text{ Hz}.$$

13.18. $T = 0.5 \text{ s}, \lambda = 10 \text{ m}, A = 0.1 \text{ m}, t = 0$ 时位移最大

$$(1) y = A \cos\left(\frac{2\pi}{T}t\right) = 0.1 \cos(4\pi t) \text{ m}.$$

距离 $\frac{\lambda}{2} \Rightarrow$ 滞后 π

$$\Rightarrow y = 0.1 \cos(4\pi t - \pi) = -0.1 \cos(4\pi t) \text{ m}$$

(2) $t = \frac{T}{2}$ 时, $x = \frac{\lambda}{4}$ 处速度为 v_0 .

$$y = 0.1 \cos\left[2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)\right]$$

$$v = \frac{\partial y}{\partial t} = -0.1 \cdot \frac{2\pi}{0.5} \sin\left[2\pi\left(\frac{t}{0.5} - \frac{x}{10}\right)\right]$$

$$t = 0.25, x = 2.5, \text{ 代入}$$

$$v = -0.4 \frac{\text{m}}{\text{s}}.$$